

UDK Phase 2A: Data Readiness + Multilingual ASR Benchmark

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1. Objective

Phase 2A makes the Phase 1 inputs usable and defines one benchmark that decides which ASR transcript representation Phase 2B builds on. It measures whether a mixed-language utterance such as "Help me, mujhe koi attack kar raha hai" survives transcription as one complete utterance, not average WER alone. No production code, no model selection by a single score, no new dataset search; sources, licences and priorities are as in the Phase 1 data strategy (§3, §7–9, §18, §20, §21, §23).

2. Data readiness table

Nothing has been downloaded or collected yet; every acquisition status is "not started". Licence status is copied from Phase 1 §20 and not upgraded here. Readiness classes: **Train** ready for training once downloaded; **Eval** ready for evaluation once downloaded; **Eval-only** never for training; **Blocked** pending licence/legal; **Collect** must be collected.

Input	Acquisition	Licence status (P1 §20)	Readiness	Preprocessing	Transcripts	Speaker meta	Language meta	Code-switch meta	Timestamps	Known limitations
IndicVoices (Hi, Te, mixed)	Not started; gated	Conditions (CC BY 4.0; gate terms unread)	Blocked until gate terms reviewed → then Train + Eval	Resample 16 kHz; derive word-level tags; CMI-based CS slice	Yes, native script	Speaker IDs	Per utterance	None; English-word script NOT VERIFIED	Utterance-level	No safety content
Vaani (transcribed)	Not started; gated	Conditions (CC BY 4.0; gate terms unread)	Blocked until gate terms reviewed → then Train + Eval	Filter transcribed clips; LID per clip	~10% of audio	Speaker metadata	Per clip	None	Clip-level	Image-description speech
Vaani Noise Event	Not started; gated	Conditions	Blocked until gate terms reviewed → then Train + Eval	Cut events by timestamps; SNR estimate	Yes, with noise tags	NOT VERIFIED	Per clip	None	Noise-event timestamps	Authors hold back 10 h eval

Input	Acquisition	Licence status (P1 §20)	Readiness	Preprocessing	Transcripts	Speaker meta	Language meta	Code-switch meta	Timestamps	Known limitations
MUCS 2021 Hi-En	Not started; open download	Conditions (CC BY-SA; share-alike under review)	Eval now; Train after legal review	Tags from script; normalise	Yes, two scripts	NOT VERIFIED	Pair per set	Script = language proxy	Sentence segments	Lecture domain
Kathbath	Not started; gated	Verified usable (CC0 packaging)	Train + Eval	m4a → wav	Yes	In file names	Per folder	None	None	Read speech
Lahaja	Not started	Conditions (CC BY 4.0)	Eval	None	Yes	Yes	Hindi	None	None	12.5 h
Svarah	Not started	Unresolved	Eval-only	None	Yes	Metadata a CSV	Indian English	None	None	Licence unresolved
IndicVoices Te-En slice (CoSTA-style)	Not started	As IndicVoices	Blocked with IndicVoices → Eval	Select by CMI; add word tags manually	Yes	Yes	Yes	Must be added	Utterance-level	~2.3 h, no tags today
FSD50K (CC0/CC-BY clips)	Not started	Conditions (per clip)	Train + Eval (noise mixes only)	Per-clip licence filter	N/A	N/A	N/A	N/A	Clip	Not Indian
AMI	Not started	Conditions (CC BY 4.0, official site)	Train + Eval (overlap)	Per-speaker alignment	Yes	Yes	English	None	Word-level	Office English
BERSt	Not started	Conditions (CC BY 4.0)	Eval (shout, distance, pocket-like)	None	Yes	Yes	English	None	Clip	Acted, nonsense phrases
Gram Vaani	Not started	Requires permission	Eval-only unless permission	mp3 → 16 kHz; note 8 kHz share	Crowd, noisy	NOT VERIFIED	Hindi	None	None	Transcript accuracy variable
NOTSOFAR-1	Not started	Unresolved	Eval-only	Device channel selection	Yes	Yes	English	None	Word-level	Licence unresolved
Phase 0 sets (prepped, real_recordings, YouTube dev, Telugu own)	Held	Own / eval	Eval-only (regression)	None	Partial	Partial	Mostly English	None	Segment-level	Tuned on (Phase 0 §13)
Collected safety speech (En, Hi, Te, Hinglish, Te-En)	Not collected	Own, consent forms under legal review	Collect	Double annotation per P1 §15 schema	Will have	Will have	Will have	Word-level tags	Full P1 §15 set	Pilot not run
Collected overlap/pocket (D_overlap)	Not collected	Own	Collect	Same	Will have	Will have	Will have	Word-level tags	Overlap spans	Pilot not run

3. Canonical manifest schema

One JSONL row per utterance, shared by every source. Fields a source cannot fill are `null`, never guessed. It extends the Phase 1 §15 annotation schema; safety fields (`pair_id`, `problem_type`, `roles`) carry over unchanged for collected data.

Field	Type	Required	Notes
<code>utt_id</code>	string	Yes	Globally unique: <code><source>:<source_id></code>
<code>audio_path</code> , <code>audio_sha256</code>	string	Yes	Hash used for near-duplicate/leakage checks
<code>source_dataset</code> , <code>source_version</code>	string	Yes	e.g. <code>mucs2021_hien</code> , release/date
<code>speaker_id</code>	string	Yes (null only if source has none → excluded from speaker-disjoint sets)	Namespaced by source
<code>session_id</code> / <code>recording_id</code>	string	If available	Groups augmented copies and multi-utterance sessions
<code>sample_rate_hz</code> , <code>duration_s</code> , <code>channels</code>	number	Yes	Original rate kept; 16 kHz copy derived
<code>language_primary</code>	string (BCP-47)	Yes	<code>hi</code> , <code>te</code> , <code>en-IN</code>
<code>languages_present</code>	list	Yes	e.g. <code>[hi, en]</code>
<code>is_code_switched</code> , <code>cmi</code>	bool, number	Yes, number if tags exist	Code-Mixing Index from tags
<code>transcript_ref</code>	string	Yes	As spoken, mixed script
<code>transcript_romanised</code>	string	If produced	For single-script text models

Field	Type	Required	Notes
tokens	list of {text, lang, script, start_s, end_s}	Where available	Word-level tags + times
script	string	Yes	Deva , Telu , Latn , mixed
english_word_script	enum	Yes for Indic sources	latin , transliterated , mixed , unknown
provenance	enum	Yes	read , extempore , conversational , acted_scripted , acted_semi_scripted , spontaneous_safe , unknown
environment	enum	Yes	quiet , traffic , crowd , home_tv , transit , office , unknown
condition_device	enum	Yes	close , phone_hand , pocket , bag , far_field , phone_call , unknown
condition_real_or_synthetic	enum	Yes	real , synthetic ; synthetic rows also name aug_recipe + seed
snr_db, clipping_ratio	number	If measurable	Estimated, method recorded
overlap, overlap_spans, overlap_rel_db	bool, list, number	Yes / if overlap	Other-speaker level relative to target
safety_tags	list	Collected + curated eval	safety_keyword , negation , short_cry , disfluency , repetition , intra_sentence_cs

Field	Type	Required	Notes
keyword_spans	list of {text, start_s, end_s}	If safety_tags	Words scored in §5
split	enum	Yes	train, dev, test, final_test, eval_only
licence, use_restriction	string, enum	Yes	Restriction: train_ok, eval_only, blocked from P1 §20

4. ASR benchmark matrix

Rows are language slices, each speaker-disjoint and reported separately. "Source" names Phase 1 inputs; nothing here is measured yet.

Slice	Sources (test)	Minimum to report	Status
Hindi	IndicVoices, Vaani test speakers; Lahaja; Kathbath (read, reference)	≥ 50 speakers, ≥ 3 h	Blocked on gate review (Lahaja, Kathbath usable)
Telugu	IndicVoices, Vaani test speakers; Kathbath	≥ 50 speakers, ≥ 3 h	Blocked on gate review (Kathbath usable)
English / Indian English	Svarah (eval-only); AMI (non-Indian control)	≥ 50 speakers	Ready once downloaded
Hinglish	MUCS Hi-En test + blind; IndicVoices high-CMI slice; collected	≥ 50 speakers, ≥ 2 h	MUCS usable for eval now
Telugu-English	IndicVoices Te-En slice with added word tags; collected	≥ 50 speakers, ≥ 2 h	Blocked: gate review + tagging

Slice	Sources (test)	Minimum to report	Status
Bengali-English (other CS)	MUCS Bn-En test + blind	Report if available	Usable for eval now
Safety-critical set (below)	Collected + curated recordings	≥ 30 speakers per P0 language	Collect

Safety-critical ASR test cases. A curated set, recorded (not TTS), where each item's full reference meaning is known. Each item is scored as one utterance; splitting it by language before scoring is not allowed, and the benchmark does not accept a pipeline that language-detects the whole utterance and routes it to one monolingual ASR as a way of passing.

Category	Example (reference)	What must survive
English embedded in Hindi	mujhe koi attack kar raha hai	"attack" + Hindi frame
Hindi embedded in English	please meri help karo, he's following me	"meri help karo" + English frame
Telugu embedded in English	help me, vaadu nannu vadalatledu	Telugu clause intact
English safety words in Indic speech	police ko call karo please	"police", "call", "please"
Full mixed sentence	Help me, mujhe koi attack kar raha hai	Every content word, both languages
Negation	mujhe help nahi chahiye / I don't need help	Negator present and in place
Short cries	"Help!", "Bachao!", "Kapaadandi!"	Word not dropped or replaced
Disfluency	h- help, please, uh, help me	Keyword recovered despite fillers
Repetition	help help help	Keyword present (count not required)

Category	Example (reference)	What must survive
Intra-sentence switch	woh mujhe follow kar raha hai	Switch point preserved

Minimum: ≥ 40 items per category per applicable P0 language, from ≥ 30 speakers. Native Hindi/Telugu wording to be written by native speakers (P1 §15); the examples above are illustrative.

Robustness subsets. Each subset is reported separately and labelled `real` or `synthetic`; a synthetic result never stands in for a real one.

Subset	Real source	Synthetic recipe (labelled)
Clean	Kathbath, IndicVoices quiet clips	—
Traffic	Vaani Noise Event (vehicle_traffic); collected street	Clean + licence-clean traffic noise
Crowd / babble	Collected market/crowd	Clean + multi-talker babble from licence-clean speech
Low SNR	Collected ≤ 0 dB	Mixes at 5, 0, -5 dB
Clipping	Collected shouted-close	Hard clip at fixed ratios
Phone-band	Gram Vaani (eval-only); collected phone_call	300–3,400 Hz band-limit + codec
Pocket / bag	Collected pocket, bag; BERSt obstructed positions	Low-pass + attenuation (as Phase 0, labelled synthetic)
Far-field	AMI far-field; BERSt distances	—
Overlapping speakers	AMI; collected D_overlap	Two-speaker mixes at 0/–6/–12 dB

5. Metrics

All metrics are computed on one normalisation: lower-case Latin, Unicode NFC for Indic scripts, punctuation removed, numbers as words. Because an English word may be written in Latin or transliterated, every text metric is computed twice — on the mixed-script reference and on the romanised copy — and both are reported.

Metric	Definition	Why it is here
WER	$(S + D + I) / \text{reference words}$	Standard; never the sole criterion
CER	Character edit rate	Fairer for agglutinative Telugu and script variation
Substitution / deletion / insertion rates	Each component of WER separately	Deletions of keywords matter more than insertions
Hallucination rate	Output words on segments with no reference speech, per hour; and insertion runs ≥ 3 words on speech	Phase 0 F3 ("Thanks for watching!")
Language-switch error rate	Share of reference tokens whose language is changed in the output (e.g. Telugu rendered as English words)	Phase 0 F4
English-word preservation (Indic speech)	Share of reference English tokens inside Indic utterances recovered as the same word, in either script	"attack", "police", "help" inside Hindi/Telugu
Negation preservation	Share of reference negators (not, don't, nahi, mat, ledu, vaddu...) present in output with correct polarity	Phase 0 F7; flipped polarity counted separately
Safety-keyword recall	Share of <code>keyword_spans</code> recovered (exact or transliteration-equivalent)	Directly tied to UDK matching
Safety-keyword false insertion	Keywords in output not in reference, per hour of harmless audio	Invented danger words drive false alarms
Code-switched utterance integrity	Share of code-switched utterances where every content word of every language is present and no language span is dropped	The "one semantic utterance" requirement

Per-utterance results are kept, not only aggregates, so failures can be listed by item.

6. Evaluation protocol

1. **Splits.** Speaker-disjoint by hashed speaker_id (P1 §18). Source-disjoint where feasible: at least one test slice per language comes from a source absent from any training or tuning used by the candidate. A candidate pretrained on a public set (e.g. IndicVoices-trained models) is scored on that set but flagged, and judged on the other sources.
2. **Frozen final test.** The benchmark's final-test portion is frozen and hashed before any candidate is run; Phase 0 sealed sets stay sealed. Candidates are compared on dev; final-test results are reported once, as totals per slice.
3. **No test-set tuning.** Normalisation rules, keyword lists and any decoding settings are fixed on dev and committed before the final-test run (Phase 0 §22 pre-registration rule).
4. **Uncertainty.** 95% confidence intervals by speaker-level bootstrap (resample speakers, not utterances), 1,000 resamples. Differences between candidates are reported with paired bootstrap intervals on the same utterances.
5. **Reporting.** Every metric per language slice × condition subset, real and synthetic separate, with item counts and speaker counts. No pooled score across languages.
6. **Baseline first.** The frozen Phase 0 engines (Whisper small.en, Parakeet TDT 0.6B v2) are run on the same benchmark before any candidate, as Phase 0 §22 requires.
7. **Latency.** Measured on CPU matching production class (c7i.2xlarge) as real-time factor per slice; T4 numbers reported only alongside.

7. Acceptance + decision criteria

Phase 2A is complete when:

- ☐ Every slice in §4 meets its minimum speakers/hours, or is explicitly marked unavailable with the reason
- ☐ The safety-critical set has ≥ 40 items per category per applicable P0 language from ≥ 30 speakers
- ☐ All robustness subsets exist with real/synthetic labels (real pocket/bag and overlap may arrive with Phase 1 collection)
- ☐ Manifest validates: no null speaker_id in speaker-disjoint splits; no speaker, audio hash or session in two splits; every row has licence + use_restriction

- ☐ Final-test portion frozen and hashed
- ☐ Frozen Phase 0 engines scored on the full benchmark as the baseline

Evidence required before Phase 2B chooses an ASR representation. No single aggregate decides. A candidate (a model, an ensemble, or an output format such as mixed-script vs romanised) is eligible only if all of these are reported, with intervals, against the baseline:

Dimension	Evidence	Rule
Multilingual accuracy	WER + CER per language slice	Reported for every P0 language; no slice hidden by averaging
Code-switch preservation	Integrity + language-switch error on Hinglish and Te-En	Must beat baseline on both CS slices
Safety-critical words	Keyword recall per category	Not worse than baseline on any category; ties broken by this before WER
Negation	Negation preservation, flips separately	No increase in polarity flips vs baseline
False danger words	Keyword false insertion + hallucination per hour	Not worse than baseline on harmless audio
Robustness	All metrics per real condition subset	Pocket and overlap reported even if weak
Latency / compute	CPU real-time factor, memory	Fits production-class CPU; not measured on T4 only
Licensing	Model and training-data licences	Commercially usable, or the candidate is excluded
Representation quality	english_word_script behaviour; romanisation round-trip	Output keeps English words recoverable

A candidate that improves WER but loses safety-keyword recall or code-switch integrity is not eligible without an explicit, recorded user decision (Phase 0 recall-first rule).

8. Open blockers

1. **Gate terms unreviewed** for IndicVoices, Vaani and Vaani Noise Event — blocks the Hindi, Telugu, Te-En and traffic slices.
2. **English-word script unknown** in IndicVoices and Vaani transcripts — blocks English-word preservation and integrity scoring on those sources until checked on a sample.
3. **No Telugu-English word-level tags exist** — the Te-En slice needs tagging of the IndicVoices slice or collected data.
4. **Safety-critical set not recorded** — needs collection (P1 §15 pilot + consent forms, which are under legal review).
5. **Real pocket/bag, crowd and overlap audio not collected** — those robustness subsets exist only in synthetic form until collection.
6. **Nothing downloaded yet** — no source has been acquired or inspected.
7. **Production CPU latency harness absent** — Phase 0 lists production CPU latency as not measured; needed for the latency criterion.

9. Inputs required for Phase 2B

- Manifest (§3) populated for every benchmark slice, with splits and final-test hash
- Benchmark results for the frozen Phase 0 engines and each candidate: per-utterance outputs, every §5 metric per slice × condition, with bootstrap intervals
- The safety-critical item list with failure cases per candidate
- CPU latency per candidate
- Licence record for each candidate model and its training data
- A written ASR-representation decision against the §7 table, including the transcript format (mixed-script, romanised, or both) Phase 2B consumes

Status

BLOCKED — the benchmark design is complete, but it cannot be run until blockers 1–6 in §8 are cleared; blocker 7 is needed before the decision in §7, not before running accuracy metrics.